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DATA DESCRIPTOR

A Comprehensive Vector Dataset of Bus Networks Across China for the Year 2024

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Urban public transportation systems facilitate traffic flow and promote sustainable urban development by offering convenient, economical, and environmentally friendly transit services. However, existing research often lacks large-scale bus network data, limiting the potential for comprehensive and in-depth analyses of urban transit systems' complexity and scope. Consequently, large-scale vector data on urban bus networks hold immense value. We designed a set of high-precision algorithms for generating bus network vector data based on electronic map APIs, which include steps such as coordinate transformation, network simplification, station projection, and topology generation. After cross-validation, a vector dataset containing bus stops, routes, and topology for 299 cities in China was generated and stored in Shapefile format (a widely used GIS file). The dataset offers valuable insights for studying network patterns in China's bus systems, supporting international comparative analyses of public transportation systems. Given that China operates the largest bus system globally, this dataset carries substantial academic and practical significance, providing data-driven insights for optimizing network structures, enhancing transportation efficiency, and informing policy-making.

Background & Summary

Public transportation is a cornerstone of modern urban operations, influencing not only the travel efficiency of urban residents but also the economic, environmental, and social development of cities. Public transportation encompasses buses, subways, light rail, ferries, taxis, shared bicycles, and electric scooters. Among these, buses are often regarded as the most critical component due to their extensive coverage, flexibility, and relatively low cost. As an indispensable part of urban public transportation, buses significantly improve residents' travel experiences, reduce traffic congestion and air pollution, promote social equity, and drive economic development.

Research on public transportation spans multiple key domains, including sustainability and environmental impacts, accessibility and equity, system robustness and resilience, optimization and scheduling, and its relationship with urban planning and design. These studies provide theoretical support and practical guidance to enhance the efficiency, sustainability, and equity of public transportation systems, thereby driving comprehensive urban traffic development. Studies have shown that bus network patterns and route design, along with the construction and selection of bus infrastructure, influence sustainable transportation and environmental outcomes^{1–4}. Furthermore, bus networks significantly affect transportation equity^{5–7}, particularly in terms of geographic accessibility^{8–10}. Recent research has focused on the robustness and resilience of bus networks, exploring how to optimize networks to prevent disruptions^{11–13}. Optimizing bus networks, routes, and stops, as well as their surrounding configurations, can improve the efficiency of urban public transportation systems^{14–18}. Bus networks, as critical urban structures, are closely tied to urban planning and land use, holding immense potential for refined transportation optimization and sustainable urban planning^{19–22}.

However, existing studies predominantly focus on single cities and are limited to specific geographic and socio-economic contexts. Comprehensive datasets covering the public transportation networks of a wide range of cities are scarce²³. Data formats and storage methods vary across cities, making it challenging to process and unify the data into standardized open datasets²⁴. Consequently, we have generated this nationwide bus network vector dataset. Nationwide datasets enable the analysis of commonalities and differences between cities, revealing patterns in sustainability, accessibility, and resilience across urban transportation systems with varying scales, development levels, and geographic distributions. Given China's vast geographical expanse, uneven

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urbanization processes, and population distribution, transportation demands and layouts vary significantly across cities and regions. Researching bus supply and demand in different areas facilitates the development of localized bus planning strategies. Additionally, China's extensive bus network and abundant data offer opportunities to uncover underlying patterns and features, aiding in optimizing resource allocation and enhancing operational efficiency.

When used in conjunction with other data, the application potential of China's bus network dataset is immense, offering new dimensions and richness to existing research. For environmental impact studies, nationwide bus route data combined with traffic flow data can support comparative analyses of carbon emissions across cities, assess the potential for electrification of bus fleets, and inform national low-carbon transportation policies. Analyzing the relationship between the built environment and bus distribution can uncover regional variations in land use changes and transportation mode choices. Regarding equity, GIS-based analyses of bus stop distributions, population densities, and socio-economic characteristics across China can identify underserved areas for vulnerable groups (e.g., low-income residents, and the elderly), guiding strategies to optimize equitable transportation services. For network structure studies, the dataset can examine the stability of various network structures and modes. Coupled with extreme weather event data, simulations of public transportation system resilience during natural disasters (e.g., typhoons, heavy rainfall) can inform strategies to enhance nationwide bus network robustness. In terms of urban and transportation planning, this dataset supports multimodal transportation planning and design, such as optimizing the integration of subways, buses, and intercity railways, and refining the spatial relationships between bus stops and other urban facilities. The dataset includes both intra-city and intercity bus routes, enabling studies on the collaborative response capacity of neighboring cities' bus networks and promoting cross-regional transportation research. Future integration with other data sources, such as census data, geographic environment data, and vehicle trajectory data²⁵, promises to enhance analysis comprehensiveness and accuracy. Machine learning models (e.g., for travel demand forecasting and optimization algorithms) could enable more precise optimization decisions. This dataset supports evaluations and improvements to national public transportation policies, contributing to urban-rural integration, carbon neutrality goals, and sustainable transportation development.

Methods

Single-source datasets are often susceptible to limitations in collection accuracy and scope, making them less comprehensive. Chinese travelers frequently use Amap (Gaode Map) and Baidu Maps for navigation and location services. Both platforms offer APIs for querying public transportation information, enabling access to bus route and station data that can be cross-validated. Using GIS technology, a vector dataset of China's bus network can be constructed. Cross-validation of multi-source data effectively enhances the validity and comprehensiveness of the dataset, facilitating more in-depth research on public transportation.

Dataset scope and units. First, we present the scope and units of our dataset. The administrative divisions of the People's Republic of China are composed of four levels: provincial-level divisions, prefecture-level divisions, county-level divisions, and township-level divisions.

Provincial-level divisions include provinces, autonomous regions, municipalities directly under the central government, and Special Administrative Regions (SARs), totaling 34 divisions: 23 provinces, 5 autonomous regions, 4 municipalities, and 2 SARs. Prefecture-level divisions consist of prefecture-level cities, regions, autonomous prefectures, and leagues, totaling 333 divisions: 293 prefecture-level cities, 7 regions, 30 autonomous prefectures, and 3 leagues. County-level divisions include urban districts, county-level cities, counties, autonomous counties, banners, autonomous banners, forest districts, and special districts, totaling 8 types. Township-level divisions include subdistricts, towns, townships, ethnic townships, ethnic subdistricts, and county districts, totaling 7 types.

Using prefecture-level cities as research units is a common practice in fields such as geography, social sciences, and economics. Prefecture-level cities have relatively complete administrative management systems and distinct geographic, economic, and cultural characteristics, making them representative of specific regions. However, considering the presence of municipalities directly under the central government and SARs, which have strong economic and cultural attributes, we also include them in our analysis. Therefore, our dataset is based on 299 cities: 293 prefecture-level cities, 4 municipalities, and 2 SARs, forming the foundation for constructing the bus network vector dataset. Due to the incompleteness of data for Sansha City, the dataset for this city is currently empty. However, to maintain the integrity of the study across all cities, we have retained its file information.

Data extraction methods and processes. The subsequent section offers a comprehensive elucidation of the methodology employed for the extraction, cross-validation, and assembly of the vector dataset pertaining to China's bus network. The process of harvesting bus data across China encompasses a series of methodical steps, which are depicted schematically in Fig. 1.

Step 1: Acquiring API keys for amap and baidu maps. The Amap APIs offer a range of interfaces, divided into Basic Web Service APIs and Advanced Web Service APIs. The Bus Information Query API, essential for our needs, falls under the Advanced Web Service APIs category. This API enables us to retrieve bus route information across various administrative regions in China.

To start, access the Amap Developer Platform, sign up for a developer account, and proceed to the console. Within the "My Applications" section, initiate a new application and generate a key, ensuring "Web Service" is selected as the platform type. This procedure yields the Amap Developer API key, a crucial credential for utilizing Amap APIs.

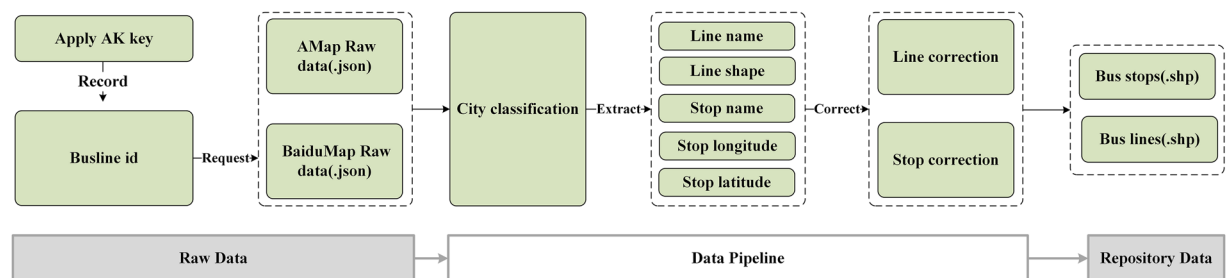


Fig. 1 Data processing pipeline for one city.

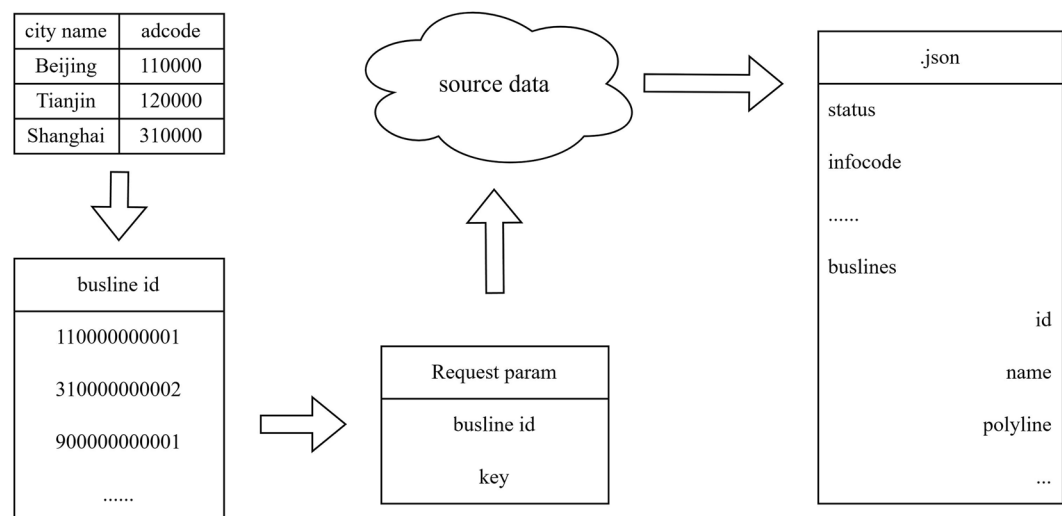


Fig. 2 Source data request.

Regarding Baidu Maps, bus route information is accessible through the JS API service. Follow a similar process by visiting the Baidu Maps Developer Platform, registering for a developer account, and entering the console. In the “My Applications” area, establish an application, choose “Browser-side” as the application type, and create a key.

Step 2: Acquiring source data. The process of acquiring source data for bus routes is divided into two components: Amap and Baidu Maps, as illustrated in Fig. 2.

The Amap Bus Information Query API provides four primary functions: Bus Stop ID Query, Bus Keyword Query, Bus Route ID Query, and Bus Route Keyword Query. Among these, the Bus Route Keyword Query presents challenges due to the difficulty of compiling a comprehensive list of route names, which may result in incomplete data. Therefore, we chose the Bus Route ID Query, which requires obtaining the unique ID for each route.

By analyzing historical Amap bus data for multiple cities, we identified the naming conventions for Amap bus route IDs. Each ID is a 12-digit number, categorized into one of two types:

1. A combination of a 6-digit city code from the Amap city code table and a 6-digit random number.
2. A combination of “900000” and a 6-digit random number.

Using the Amap city code table, we performed an iterative traversal to retrieve all bus route IDs for the cities within our research scope. With the Amap API key obtained in Step 1, we constructed request URLs and sent requests using Python’s requests library. By parsing the returned data, we obtained JSON-formatted bus route source data. The extracted data structure, a sample of which is shown in the table below, includes vector information such as route names, geometric coordinates (longitude and latitude), stop names, and stop coordinates. For each city, we saved the route data in two separate “CSV” files: one for bus routes and another for bus stops. Additional details on Amap Web API parameters can be found in the developer documentation: [Amap API Documentation](#).

The acquisition of source data for Baidu Maps primarily leverages the “BusLineSearch” class and its associated methods provided by the JavaScript API. Baidu Maps permits the retrieval of bus route information solely based on route name keywords. Given that a comprehensive list of route names has already been obtained from Amap, this list can be utilized to query Baidu Maps. In the JavaScript code, a “BusLineSearch” object

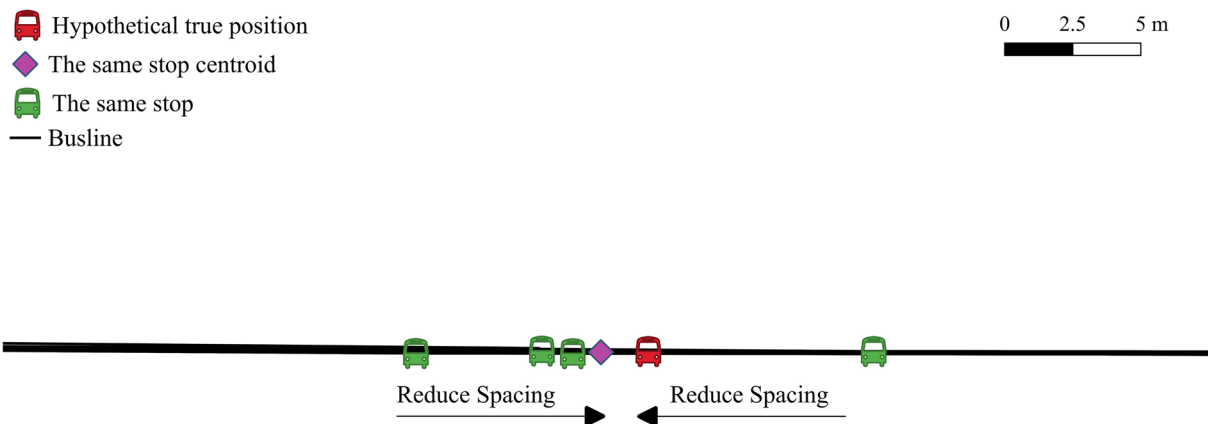


Fig. 3 Stop simplification.

is instantiated. By employing its “getBusList” and “getBusLine” methods, bus route information is iteratively retrieved from Baidu Maps using the provided keywords. Similar to the data processing for Amap, for each city, the route data is stored in two separate “.CSV” files: one for bus routes and another for bus stops. For detailed information regarding the parameters of the Baidu Maps JavaScript API, please refer to the official developer documentation: [Baidu API Documentation](#).

Step 3: Data simplification. In Step 2, we acquired bus route and bus stop data files (.CSV) for each city from both Amap and Baidu Maps. In this step, we focus on data validation, cross-verification, and simplification to ensure consistency and usability of the dataset.

To ensure the universality and compatibility of the vector dataset, it is essential to address the differences in coordinate systems used by Amap and Baidu Maps. Here’s a refined explanation of the process, i.e., coordinate system conversion. Amap uses the GCJ-02 coordinate system, which is an encrypted version of the standard WGS-84 geographic coordinate system. This encryption introduces slight deviations, making it incompatible with global datasets that rely on WGS-84. Baidu Maps employs the BD-09 coordinate system, which adds an additional layer of encryption on top of GCJ-02. This further complicates the integration of Baidu data with other geographic datasets. To ensure the universality of the vector dataset, both GCJ-02 (Amap) and BD-09 (Baidu Maps) data must be converted to the WGS-84 coordinate system. This conversion aligns the data with the global standard, enabling seamless integration with other datasets and tools.

The station data presents a unique challenge due to the nature of its collection. Specifically, the vector data for routes and stations are similar to the GPS trajectory records of buses. Considering the complex station-setting rules of China’s public transportation system, there are many bus routes in certain urban areas. Bus stations with the same name may be located in adjacent spatial positions. Therefore, although their names are the same, their latitude and longitude coordinates have numerical differences. In the study of public transport networks, these stations are usually considered as a single station due to their identical names and adjacent locations. This is the motivation for the simplification of station locations in this paper. Meanwhile, a small part of the reason comes from the collection errors of the same station location by the two map service providers, Amap and Baidu Maps. As a result, when multiple routes pass through the same station, each route may record slightly different vector coordinates for that station. This leads to minor discrepancies in the recorded positions of stations. To address this issue and eliminate duplicate data, we employed a method to standardize the position of each station. We calculated the centroid of all recorded coordinates for each station and used this centroid as the standardized location for that station. This approach effectively consolidates the data, ensuring consistency across all routes. The process is illustrated in Fig. 3.

For the verification of route data, limitations in data collection scope, accuracy, and cost may result in varying focuses across different map developers, potentially leading to omissions in bus route data. To address this, we compare the two source datasets and perform cross-validation. By selecting the dataset with more complete route information and supplementing any missing details from the other source, we aim to create a comprehensive and accurate set of route data.

Step 4: Construction of simplified bus network dataset. In Step 3, we obtained the unified bus route and bus stop files (.CSV) for each city after validation and cross-verification. In this step, we convert these files into the Shapefile vector format, a widely used geospatial data format, using the “Geopandas” library in Python. Below is a detailed breakdown of the process. Using Python, we first create a new GeoDataFrame for the bus routes, including columns for “lineName” and “geometry”. For each route’s coordinate data, we utilize the “shapely” library to generate a LineString object, which is subsequently assigned as the geometry data. The route name is stored in the “lineName” column. Finally, we save the data as a Shapefile using the “.to_file” method.

For bus stops, we follow a similar process by creating a new GeoDataFrame using the “Geopandas” library, with columns for “stopName”, “lineName” and “geometry”. For each stop’s coordinates, we use the shapely library to create a Point object, which is then assigned as the geometry data. The name of the stop is stored in the

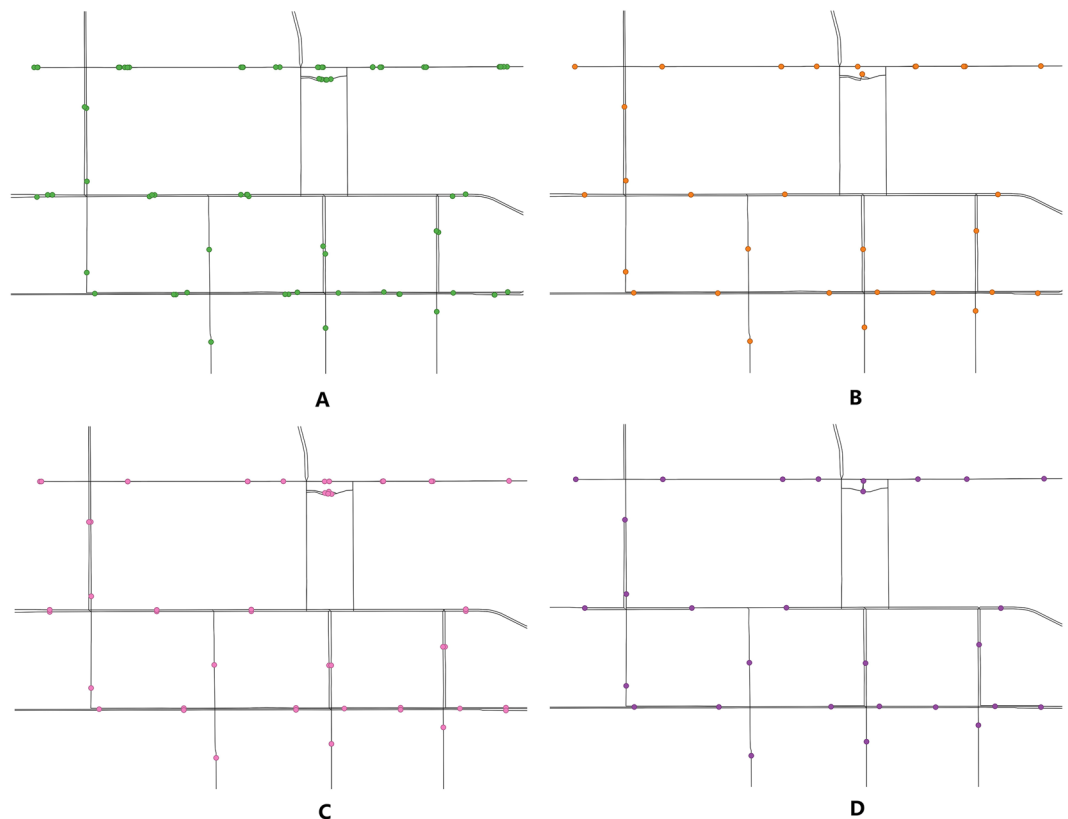


Fig. 4 The different stages of network construction. (A) Original network. (B) Simplified network. (C) Stop projection. (D) Topological network.

“stopName” column. Finally, the data is saved as a Shapefile using the “.to_file” method, ensuring the bus stop information is stored in a standardized geospatial format for further analysis or visualization.

Step 5: Construction of topological bus network dataset. Following Step 4, we constructed a simplified urban bus network dataset containing geographic information on routes and stops. To enable network analysis, we further developed an undirected topological graph using the L-Space model²⁶. This model is well-suited for bus networks, as it treats bus stops as nodes and the routes between stops as edges, aligning with the operational logic of bus systems.

However, since bus route trajectories are derived from data similar to GPS data, routes on the same road may not fully overlap. To address this, we: (1) projected bus stops onto routes and segmented the routes based on stop locations to establish connections between stops. (2) retained only one edge for node-to-node relationships shared by multiple routes, avoiding redundancy. (3) handled non-overlapping trajectories by calculating the centroid of identical nodes and extending edges to connect to this centroid, ensuring the topological graph remains fully connected. This process ensures the topological graph accurately represents the bus network’s structure, enabling robust network analysis and research.

The complete process for producing the Chinese bus network dataset is shown in Fig. 4.

Data Records

The vector dataset of the bus network in China (data for Taiwan Province is not available) up to April 2024, formatted in Shapefile, is available on Figshare²⁷. The original language of this dataset is Chinese, and we have used the Microsoft Translation API to uniformly translate the Chinese content into English.

For each city, the dataset contains two types of data: bus stops and bus routes (distinguishing direction, with one being undirected and the other being directed). The topological network dataset consists of nodes and edges, with the edges being undirected.

A Shapefile consists of multiple files, each serving a specific purpose:

- (.shp): This is the main file containing geometric data such as points, lines, and polygons. All coordinate information is stored here.
- (.dbf): This file contains attribute data, with each record corresponding to a geometric object in the “.shp” file.
- (.shx): This file stores index information for the geometric data in the “.shp” file, enhancing search efficiency.
- (.cpg): This file specifies the character encoding used in the “.dbf” file.

Shapefile	Field	Description
busline	lineName	Route ID, the starting and terminal stops
	geometry	Shapefile line shape data
busstop	stopName	Bus stop name
	lineName	The route number to which the station belongs and the first and last stations.
	geometry	Shapefile point shape data

Table 1. Field of each table in the database.

City	Stops		Routes	
	OSM	Our Dataset	Official statistics (the end of 2023)	Our Dataset (April, 2024)
Shanghai	10312	16313	1589	1808
Beijing	8258	15685	2132	2352
Guangzhou	6591	11284	1369	1747
Shenzhen	4700	7113	942	1326
Chongqing	790	14409	791 (central area)	1939
Suzhou	1229	12182	1269	1412
Chengdu	461	14072	1422	1757
Hangzhou	2046	12603	1495	1816
Wuhan	625	8231	592	861
Nanjing	1515	8449	823	950

Table 2. Comparison of the number of bus stops and routes in different datasets.

This dataset can be visualized and edited using software such as ArcGIS Pro, ArcMap, QGIS, or programming libraries like Geopandas and Shapely in Python.

Table 1 provides an overview of the data attribute fields for the simplified bus network dataset in Shapefile format.

Technical Validation

The validation process encompasses two main components: a comparison with OpenStreetMap data and a statistical analysis of our dataset to elucidate the characteristics of public transportation in China.

We selected the top ten cities in China by GDP and conducted a detailed comparison between the number of bus stops in the OSM dataset and those in our dataset. The two types of data have been processed to remove duplicates, retaining only one instance for each station name. As illustrated in Table 2, our dataset consistently exhibits a higher number of bus stops compared to the OSM dataset. Notably, exception for Shanghai, Beijing, Guangzhou, and Shenzhen, the OSM dataset reveals significant gaps in coverage for the remaining cities, capturing only a limited number of bus stops primarily located in the central urban areas. The main reason is that OpenStreetMap is restricted in China, and Chinese users can only access and edit it through a VPN. Many users are accustomed to using well-established commercial maps (such as Amap and Baidu Maps), which makes them less likely to pay attention to and participate in the updating and maintenance of open-source map data, resulting in low data update activity of OSM in China.

Meanwhile, the number of routes we collected also exceeds the official statistical information. One reason is that the official bus data statistics are lagging, and the data is not updated in a timely manner. Our data was updated in April 2024, which is later than the official statistics at the end of 2023. Another reason is that the scope of official bus statistics is limited, usually focusing mainly on data within the developed urban areas, while our dataset includes routes across the entire administrative region. Moreover, the public transportation systems in Chinese cities are generally operated jointly by multiple companies, and not all companies disclose relevant data. For example, in Hangzhou, we only collected bus information displayed by the Hangzhou Public Transport Group, while other companies, such as the Yuhang Public Transport Co., Ltd. in Hangzhou, did not display any information.

Leveraging this bus vector dataset, we performed a statistical analysis to investigate the public transportation characteristics of Chinese cities. We specifically selected cities with at least 20 bus routes to ensure sufficient data for meaningful analysis, which led to the inclusion of 289 cities in total. The bus-related indicators chosen for the statistical analysis are presented in Table 3.

As indicated in Table 3, the standard deviations of both the number of bus routes and the total length of bus routes exceed their respective means. This suggests substantial variability in the number and length of bus routes across different prefecture-level administrative districts in mainland China. Such variation is likely influenced by differences in city size, which significantly impact the scale of bus networks. In contrast, the standard deviations of indicators such as average bus route length, average number of bus stops per route, average distance between stops, bus network density, bus network redundancy coefficient, and average non-linearity coefficient of bus routes are relatively small, indicating low variability. The average non-linearity coefficient of bus routes is 1.74, which is considered relatively high. This suggests that bus routes in various cities tend to be more winding, possibly due to factors such as natural terrain obstacles and historical planning constraints.

Indicator	Minimum	Maximum	Mean	Median	Standard Deviation
Bus routes	20	2473	274	159	361
Total length (km)	235.56	64676.99	6159.67	3623.18	7424.85
Avg route length (km)	9.81	67.81	23.22	22.31	7.71
Avg route stops	11.08	47.09	24.11	24.19	4.66
Avg stop distance (m)	488.45	3400.34	1026.02	941.81	362.58
Density (km/km ²)	0.50	4.17	1.86	1.90	0.45
Redundancy coeff	1.83	29.22	4.06	3.79	1.98
Non-linearity coeff	1.35	2.35	1.74	1.73	0.16
500 m coverage (%)	0.21	0.94	0.76	0.79	0.13

Table 3. Descriptive statistics analysis.

Code availability

The code used for extracting and processing data was released on Figshare with bus data²⁷. Additionally, the parameters for data requests, such as the rules for bus line IDs, or commercial service availability may change. Amap's commercial services are available on the URL, however, Baidu Maps had suspended this service when it came to March 2025.

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Author contributions

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Competing interests

The authors declare no competing interests.

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